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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738153057521-Chapter 8 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

INTRODUCTION

In Unit 2, you'll take what you've learned about lung health and spirometry and start creating your own spirometer. This device will measure lung capacity and breathing efficiency. You'll plan how the spirometer will work, choose the right MODI modules, and begin building the device that can help monitor lung health effectively.

IDEA GENERATION: STEPS TO GET STARTED

The Smart Posture Detector we are going to build will use sensors to monitor your distance from the monitor and give you feedback when you are too close. Here's how it works:

1. IDENTIFY THE CORE FUNCTION

Determine the main purpose of your spirometer: measuring lung capacity accurately. Consider what tasks your device needs to perform to achieve this goal.

2. BRAINSTORM COMPONENTS AND DESIGN

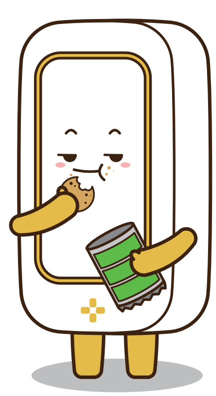
Think about which MODI modules and materials you'll need. Sketch an initial design to visualize how these components will be arranged and connected.

3. PLAN USER INTERACTION AND DATA HANDLING

Consider how the user will operate the device, including starting and stopping the test. Plan how your spirometer will display and handle the data it collects.

TIPS

Keep your design user-friendly! Think about how someone unfamiliar with your device would use it.
Clear instructions and easy-to-understand displays are key.

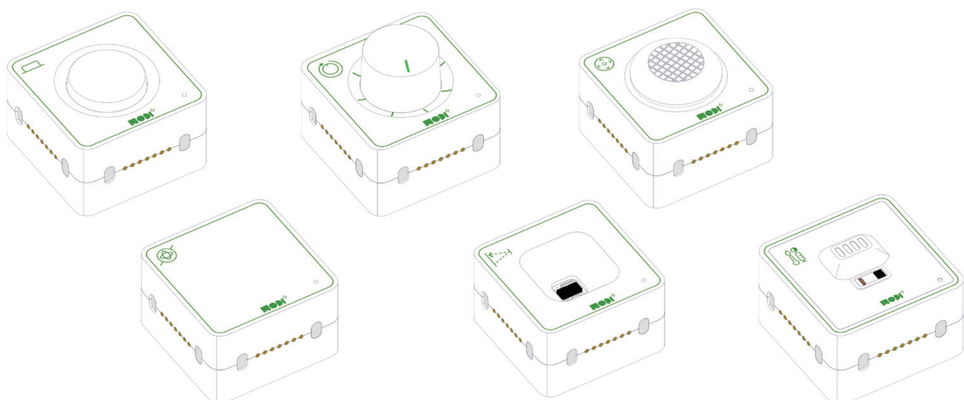


Required Modules & Brainstorming

To build our Spirometer, we need to decide which MODI modules to use. Here are some ideas to get started:

SENSOR MODULES

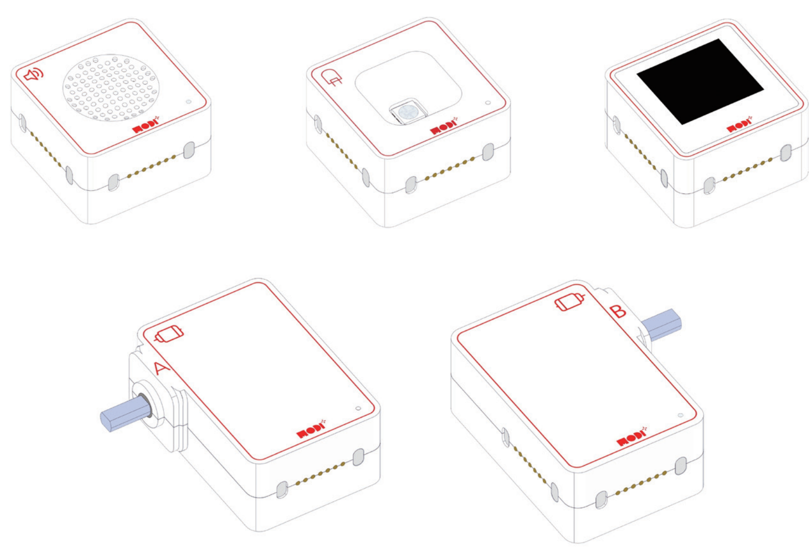
Detecting Air Flow: Your spirometer needs a sensor that can measure how much air is inhaled and exhaled. Think about which sensor could detect changes in air pressure or flow when you breathe into the device.



OUTPUT MODULES

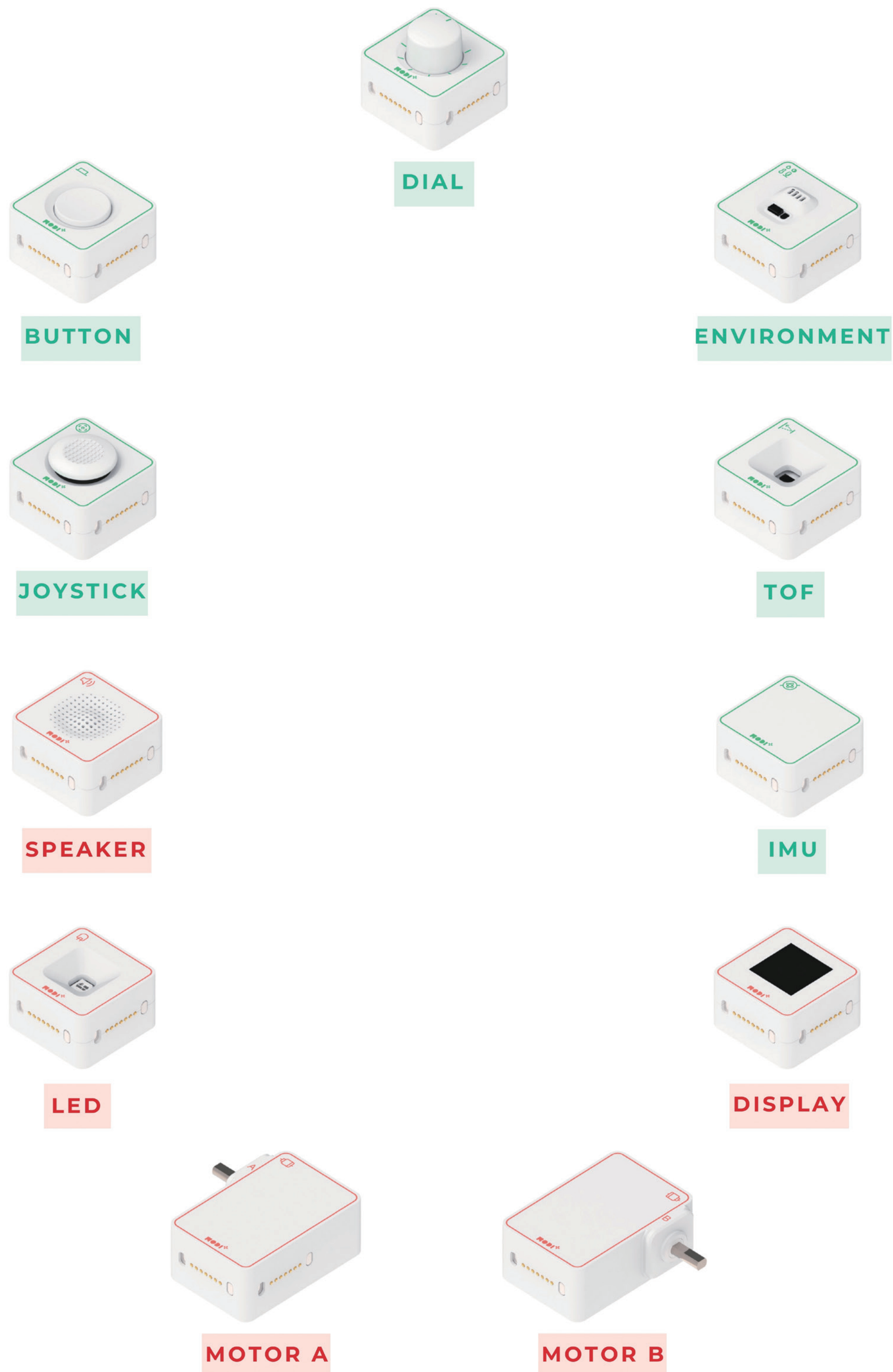
Displaying Results: After measuring lung capacity, your spirometer needs to show the results to the user. Consider what type of output would be best for displaying numerical data or messages.

Visual Feedback: It would be helpful if your spirometer could give immediate visual cues during the test. Think about how a simple light or display could indicate when to start or stop exhaling, or to signal the completion of the test.

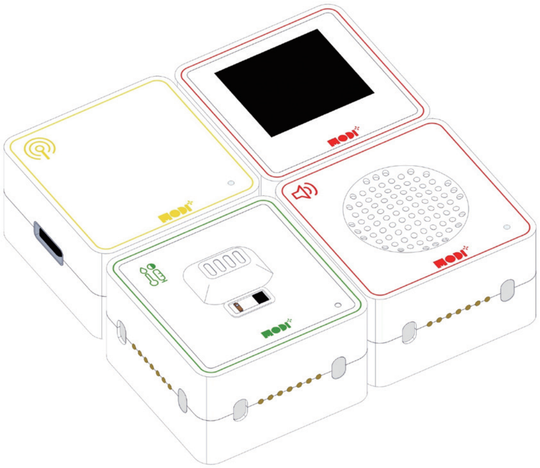


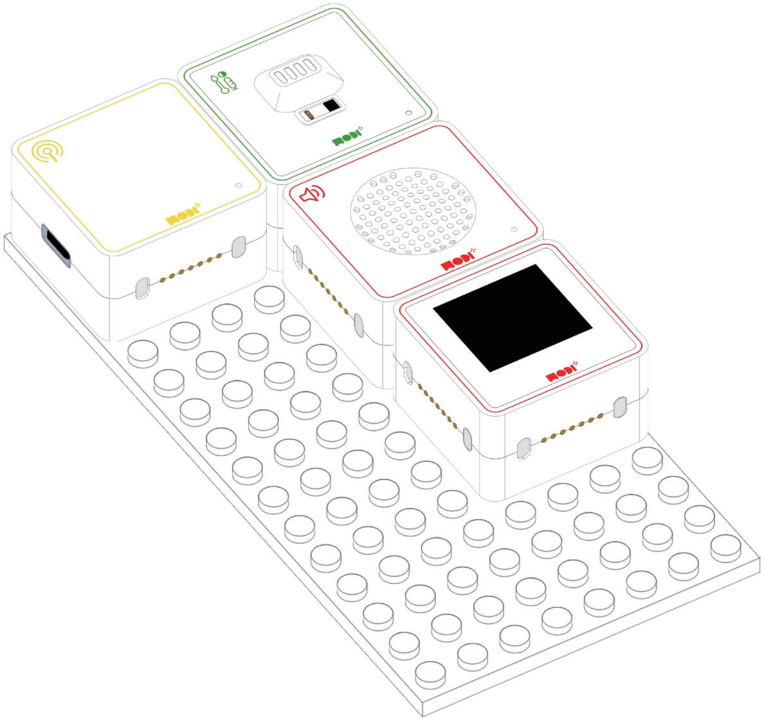
Required Modules & Brainstorming

Which MODI modules do you need for the creation?



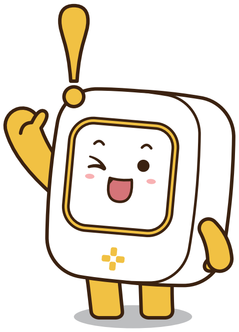
WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.





TIPS

MODI modules can be connected in any direction! This makes building your projects easier and more flexible. No need to worry about which way they face! That;s your choice!



Module Functionality

ENVIRONMENT SENSOR

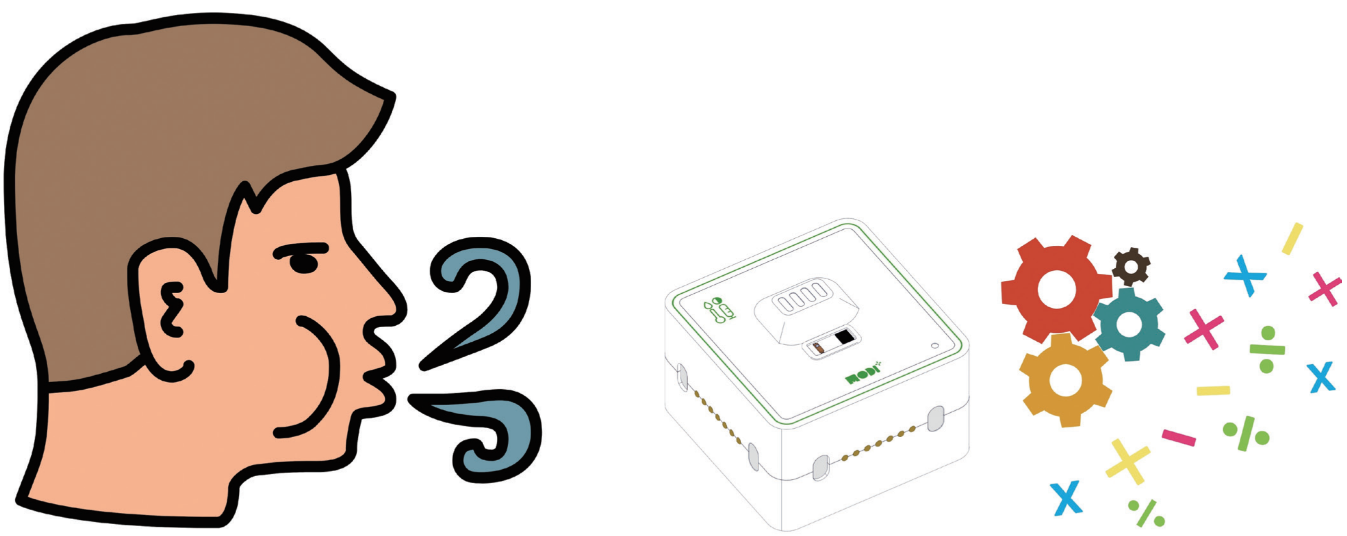
The Environment Module in your spirometer is equipped with a sound level detector, which it uses to measure air flow. When you blow into the spirometer, the sound created by the airflow is detected by the module. This sound data is then used to estimate the volume of air being exhaled, which is crucial for calculating lung capacity.

Hi~

How the Environment Module Works

KEY FUNCTIONALITY

- **Sound Detection:** As you exhale into the spirometer, the Environment Module picks up the sound produced by the airflow. The louder and more consistent the sound, the higher the air flow rate, which directly correlates to the volume of air being expelled from the lungs.
- **Data Conversion:** The module converts the detected sound levels into data that can be processed by the system. This data represents the air flow and is crucial for determining lung capacity.
- **Real-Time Monitoring:** The Environment Module continuously monitors the sound levels during the spirometry test, providing real-time data on the air flow. This allows the spirometer to give immediate feedback, such as displaying the results or triggering visual cues through an LED module.



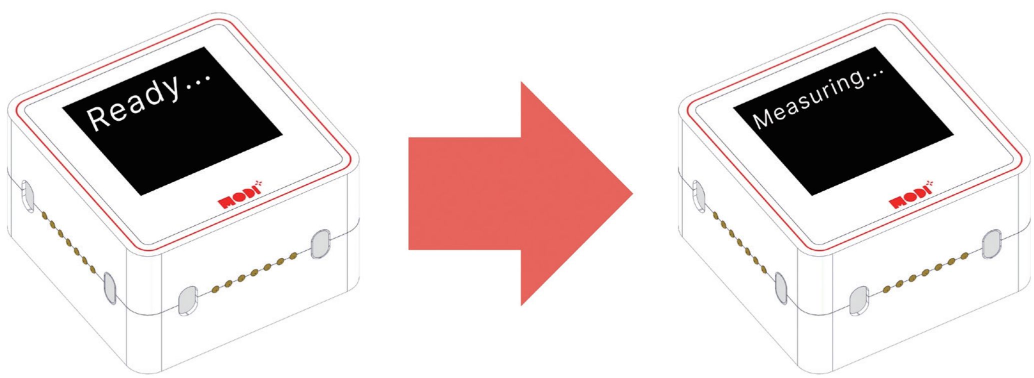
Duration of Sound =
How long a person can exhale their breath

USES OF THE DISPLAY MODULE

The display module in this project is primarily used to communicate information to the user about the status of the spirometer and the results of the breathing test. Here's how the display module is utilized:

1. STATUS UPDATES:

- The display shows the initial status as "Ready..." indicating that the spirometer is prepared for the user to start the test.
- During the test, when the environment sensor detects sound (indicating air being blown into it), the display changes to "Measuring..." to inform the user that the device is actively recording the duration of the exhale.



2. RESULTS DISPLAY:

- After the test is completed, the display shows the result of the measurement (stored in the variable record). This variable reflects the duration of the exhale, which is a key metric in determining lung health.
- The result is displayed on the first line of the display module, giving immediate feedback to the user.